DEPI GRADUATION PROJECT

Task 1 - Project Planning & Management

Automated Teacher Workflow with n8n

DEPI Recommended Capstone Project #9 - AI in Education

Track	AI in Education
Team Size	5 Members
Team Members	• Heba Abdelhameed Alaa Eldeen Abdelhameed Bakr • Hend Ahmed Hassan Sayed Ahmed • Eman Mahmoud Siddik Awad • Marwa Ahmed Mohamed • Fatma Mohamed Ahmed
Group Code	GIZ5_IND1_G1
Task Deadline	2 Oct 2026

1. Project Proposal

Project Name: Automated Teacher Workflow with n8n
Project Type: AI-enabled educational automation workflow
Primary User: Teachers and teaching professionals

1.1 Project Overview

The Automated Teacher Workflow with n8n is an AI-enabled automation project that helps teachers reduce repetitive instructional content-production tasks. The teacher submits an instructional topic and related lesson information, and an n8n workflow automatically passes the data through multiple AI-powered steps to generate a lesson outline, student activities, an assessment, an answer key, and a differentiated version. The final outputs are organized for teacher review before use.

The project focuses on workflow automation rather than building a full learning platform. It demonstrates triggers, multiple workflow nodes, data passing, AI transformation, branching, conditions, error handling, a fallback path, structured output formatting, and documentation.

1.2 Problem Statement

Teachers repeatedly perform administrative and instructional content-production tasks that can be partially automated. Creating lesson outlines, activities, assessments, answer keys, and differentiated versions separately can consume time and create unnecessary repetition.

The project addresses this problem by creating one automated workflow in which the teacher provides the instructional input once, and the system processes it through a sequence of connected steps to produce multiple organized outputs for teacher review.

1.3 Project Objectives

- Automate a sequence of repetitive teacher content-production tasks using n8n.
- Accept a single instructional input and reuse it across multiple workflow steps.
- Generate a lesson outline, student activities, assessment, answer key, and differentiated version.
- Demonstrate data passing between connected workflow nodes.
- Apply AI transformation through structured prompts at relevant workflow stages.
- Use branching and conditions to control workflow behavior.
- Include input validation, error handling, and a fallback path.
- Format the final outputs in a clear structure for teacher review.
- Document the workflow, prompts, test cases, and error/fallback behavior.

1.4 Project Scope

Proposed Inputs

- Subject
- Grade level
- Topic
- Learning objective
- Lesson duration (when relevant)
- Learner level or differentiation preference (when relevant)

Core Workflow Outputs

- Lesson outline
- Student activities
- Assessment
- Answer key
- Differentiated version
- Organized final output for teacher review

Core Technical Scope

- Trigger
- Multiple workflow nodes
- Data passing between nodes
- AI transformation
- Branching
- Conditions
- Error handling
- Fallback path
- Output formatting
- Documentation

Optional Extension

- Connect the workflow to an external API or educational platform for automatic data retrieval or submission. This extension will only be considered after the core workflow is complete.

1.5 Proposed High-Level Workflow

1. Teacher submits the instructional topic and required lesson information.
2. n8n trigger starts the workflow.
3. Input validation checks that required information is available.
4. AI step generates the lesson outline.
5. AI step generates student activities.
6. AI step generates the assessment.
7. AI step generates the answer key.
8. A condition/branch applies the required differentiated version.
9. The workflow formats and combines the outputs.
10. Teacher receives the organized output for review.
11. If a node fails or required data is missing, the workflow follows an error/fallback path.

1.6 Technical Approach Based on Course Learning

Implementation will use structured inputs, prompts, connected nodes, validation, testing, and iterative refinement, reflecting the workflow concepts covered in the track.

2. Project Plan

The following plan follows the official DEPI milestone dates and divides the work into short implementation phases.

Period	Phase	Main Activities	Deliverable	Owner(s)
1-2 Oct	Planning & Management	Finalize proposal, scope, timeline, roles, risks, KPIs, GitHub structure, and workflow concept.	Task 1 documentation	All team / Team Leader
3-5 Oct	Requirements & Research	Stakeholder analysis, user stories, use cases, functional/non-functi onal requirements, and literature review.	Requirements + literature draft	Members 1, 2, 5
6-9 Oct	System Analysis & Design	Create workflow diagram; define nodes, data flow, conditions, branches, prompts, outputs, and error/fallback logic.	System design documentation	Members 2, 3, 4
10-13 Oct	Implementation	Build n8n workflow, configure trigger, connect AI steps, pass data between nodes, implement branching and output formatting.	Working n8n prototype	Members 2, 3
14-15 Oct	Testing & Refinement	Run test cases, test invalid inputs and node failures, refine prompts, fix workflow issues, and document fallback behavior.	Test evidence + refined workflow	Members 3, 4, 5
16 Oct	Final Delivery	Complete documentation, README, workflow export, final presentation, reports, and project demo.	Final submission package	All team

2.1 Gantt Overview

Phase	1-2 Oct	3-5 Oct	6-9 Oct	10-13 Oct	14-15 Oct	16 Oct
Planning & Management	●					
Requirements & Research		●				
System Analysis & Design			●			
Implementation				●		
Testing & Refinement					●	
Final Delivery						●

2.2 Milestones

- M1 - Project planning completed and responsibilities assigned.
- M2 - Requirements, research, and workflow design completed.
- M3 - Core n8n workflow runs end-to-end for a valid teacher input.
- M4 - Branching, conditions, error handling, and fallback path tested.
- M5 - Documentation, test evidence, final presentation, and demo completed.

2.3 Deliverables

- Project proposal and project plan
- Task assignment and role definition
- Risk assessment and mitigation plan
- KPI definition
- Requirements and system-design documentation
- Working n8n workflow
- Workflow diagram
- AI prompts
- Test cases
- Error/fallback demonstration
- Final documentation, presentation, and demo

2.4 Resource Allocation

The project team consists of five members. Each member has a primary area of responsibility, while workflow review, testing, and final decisions are shared across the team.

3. Task Assignment & Roles

Team Member	Primary Role	Main Responsibilities	Key Outputs
Member 1	Team Leader & Project Coordination	Track milestones and deadlines; coordinate tasks; communicate with the instructor; manage GitHub organization; consolidate deliverables.	Project coordination, milestone tracking, integrated submission
Member 2	Workflow Architecture & n8n Development	Design the workflow structure; configure trigger and nodes; manage data passing; support branching, conditions, and integrations.	Workflow diagram and core n8n workflow
Member 3	AI Prompt & Workflow Logic	Write and refine AI prompts; define transformation steps; support branching logic and structured output generation.	Prompt set, AI transformations, output logic
Member 4	Error Handling, Testing & QA	Prepare test cases; test normal and failure scenarios; verify error handling and fallback behavior; log issues and resolutions.	Test cases, QA evidence, error/fallback demonstration
Member 5	Documentation & Final Output	Prepare project documentation; organize final teacher-facing output; support requirements, README, final presentation, and demo.	Documentation, formatted outputs, presentation/demo support

4. Risk Assessment & Mitigation Plan

Risk	Potential Impact	Mitigation Plan	Primary Owner
Missing or unclear teacher input	The workflow may generate incomplete or unsuitable outputs.	Validate required fields before AI generation and route incomplete inputs to a clarification/fallback path.	Members 2 & 3
AI output is irrelevant or inconsistent	Generated materials may not match the topic or intended lesson context.	Use structured prompts, pass the same core context between nodes, and refine prompts through repeated testing.	Members 3 & 4
n8n node or integration failure	The workflow may stop before producing the final output.	Test nodes independently, use error handling, and define retry/fallback behavior.	Members 2 & 4
Branching/condition logic behaves incorrectly	The wrong workflow path or output may be generated.	Define clear conditions and create dedicated test cases for every branch.	Members 2 & 4
API/model limit or temporary service failure	AI steps may fail or become temporarily unavailable.	Monitor usage limits, document retry behavior, and keep a supported alternative configuration where possible.	Members 1 & 2
Final outputs are poorly organized	Teacher may find the result difficult to review or use.	Use a consistent output template and review formatting in end-to-end tests.	Members 3 & 5
Scope becomes larger than available time	Core workflow may remain incomplete.	Prioritize the official Project 9 core scope and postpone external API/platform integration until the core workflow works.	Team Leader
Team work is not integrated regularly	Late integration may create conflicts or missing documentation.	Use one GitHub repository, assign clear ownership, and review progress at short milestones.	All team

5. Key Performance Indicators (KPIs)

KPI	How It Will Be Measured	Initial Target	Purpose
End-to-End Workflow Success Rate	Successful complete runs divided by total valid-input test runs.	≥ 90%	Measures overall workflow reliability.
Required Output Completion	Number of required outputs produced in a successful run.	100% of core outputs	Confirms that the workflow completes the intended automation sequence.
Branching & Condition Accuracy	Correct branch selected in predefined branch test cases.	100% of defined branch cases	Confirms workflow logic is functioning correctly.
Error/Fallback Handling	Defined failure and invalid-input cases handled without producing a misleading final result.	100% of defined error cases	Measures robustness of failure handling.
Content Relevance	Team/teacher-review checklist comparing generated outputs with topic, grade, objective, and input constraints.	≥ 85% acceptable results in test set	Checks usefulness and alignment of AI outputs.
Output Formatting Completeness	Final package follows the agreed output structure with all sections clearly labeled.	100% of successful runs	Ensures teacher-facing outputs are organized and reviewable.
Processing Time Stability	Record total run time across repeated tests and compare against the established test baseline.	No unexplained major increase across repeated tests	Tracks workflow efficiency without assuming a fixed API response time.

6. Project Assumptions and Boundaries

- The project will focus on a working n8n automation and will not require building a full educational platform.
- The teacher remains responsible for reviewing AI-generated materials before classroom use.
- External API or educational-platform integration is treated as an optional extension, not a core dependency.
- The team will use one GitHub repository for project files, documentation, workflow exports, prompts, and updates.
- Any technical change that significantly expands the official Project 9 scope will be discussed with the technical instructor first.